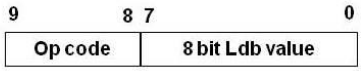
Mini24 CPU Hardware Design

By: Noam Hayu & Charbel Marshi

Instruction	Symbolic Description	Description
and	A←A and B	Logic "and"
add	A←A + B	Arithmetic sum
ldb <8 bit number>	B←8 bit number	Load an 8 bit number to B register
Hlt		Stop CPU

OP code
and 00
add 01
Ldb 10
hlt 11



Control Logic Equations

- AR (counter)

  $inc = \neg D_3 \cdot T_0$

- SC (counter)

  $Ld = T_1$

  $inc = \neg D_3 \cdot T_0$

- IR Register

  $Ld = \neg D_3 \cdot T_0$

- A Register

  $Ld = (D_0 + D_1) \cdot T_1$

- B Register

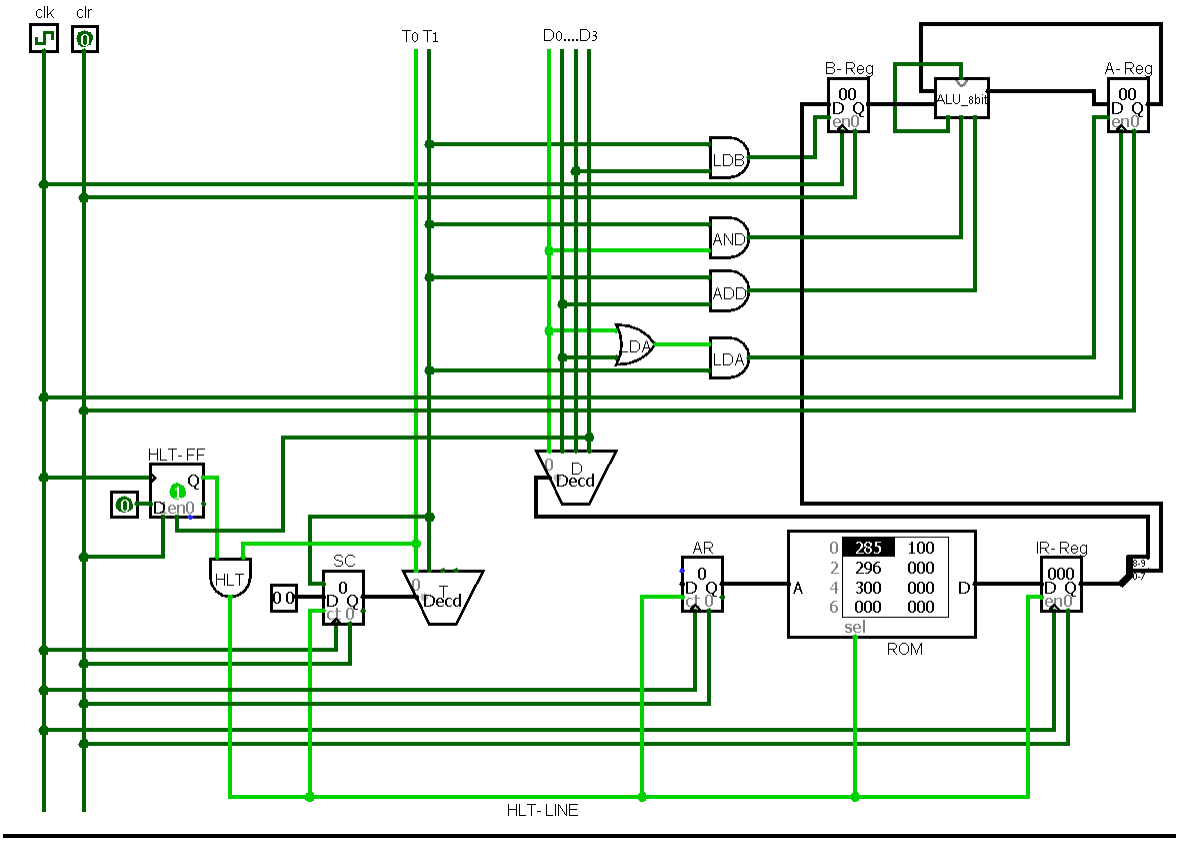
  $Ld = D_3 \cdot T_1$

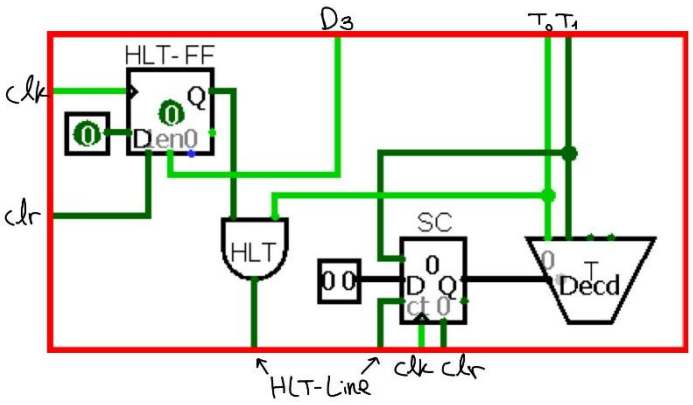
- ALU

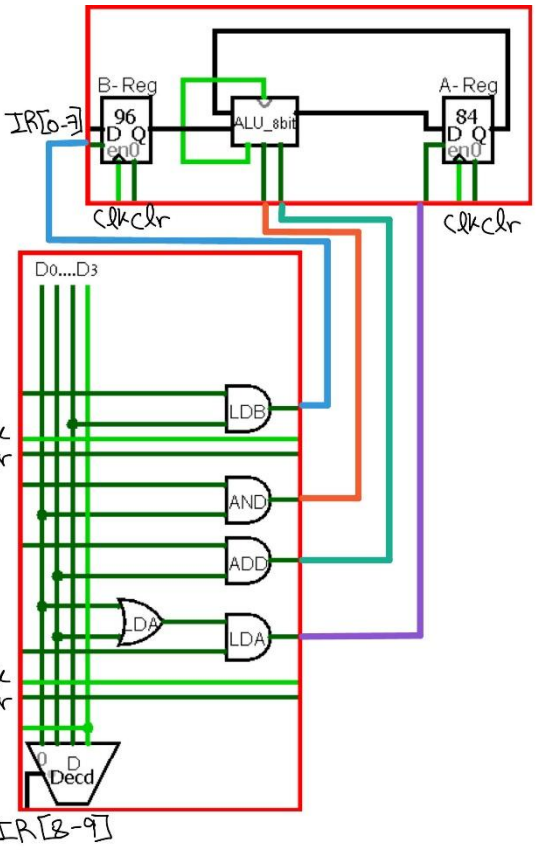
  $ADD = D_1 \cdot T_1$

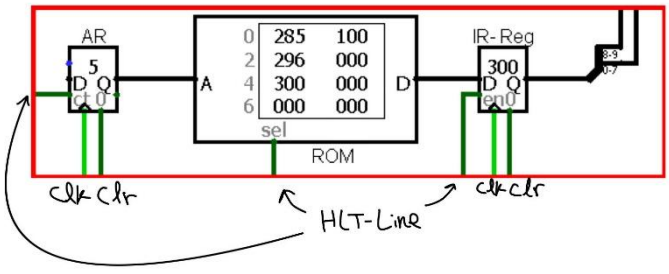
  $AND = D_0 \cdot T_1$

Logisim Circuit Diagrams





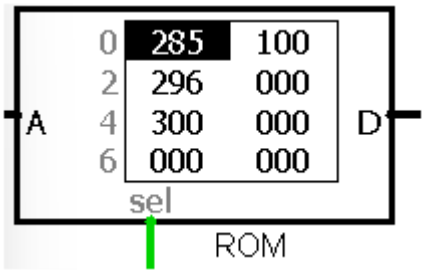




Machine Code Encoding

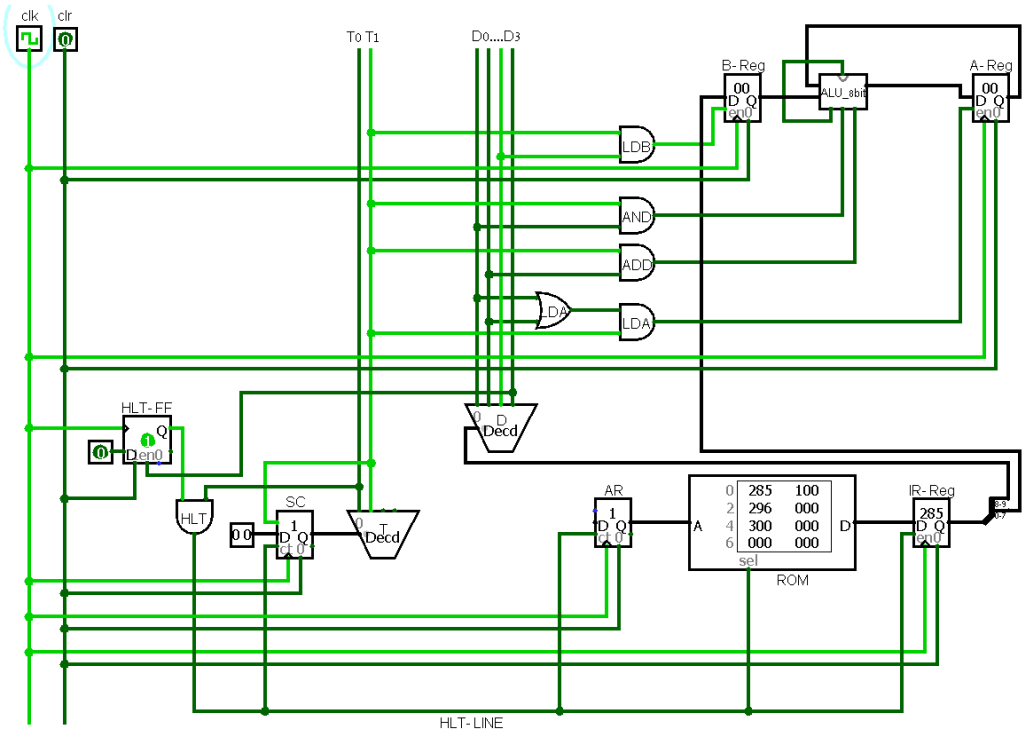
Assembly	Hexadecimal code	Binary code
ldb 85h	285	1010000101
add	100	0100000000
ldb 96h	296	1010010110
and	000	0000000000
hlt	300	1100000000

ROM Contents

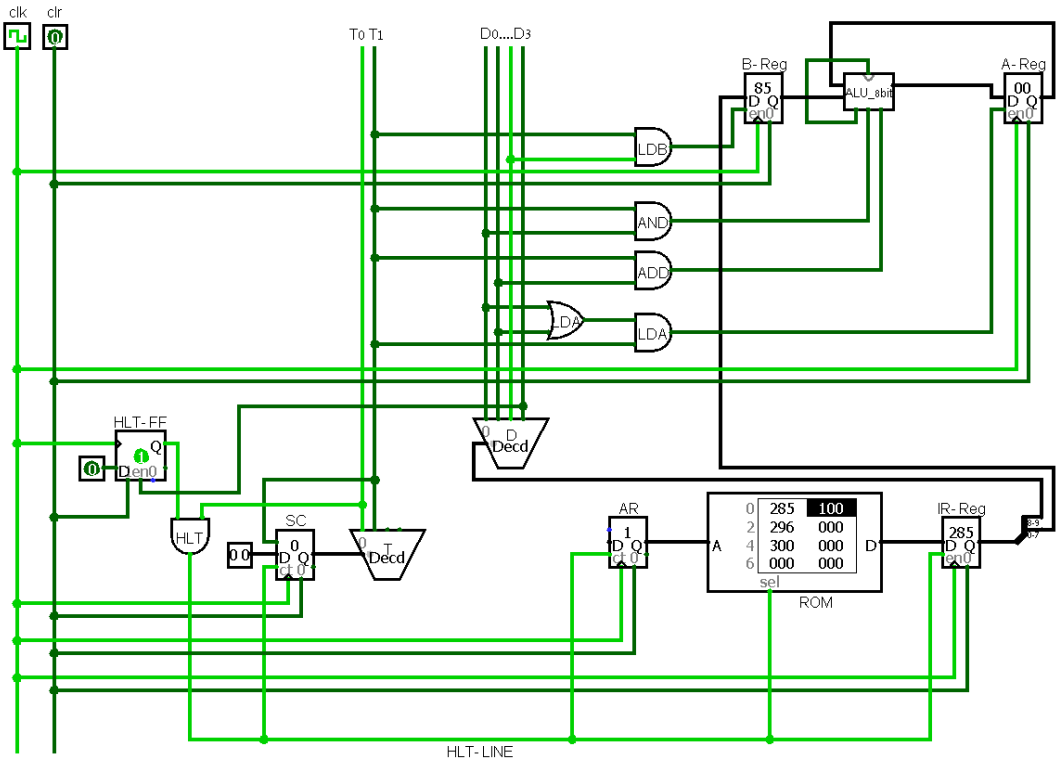


Execution Trace (Step-by-Step)

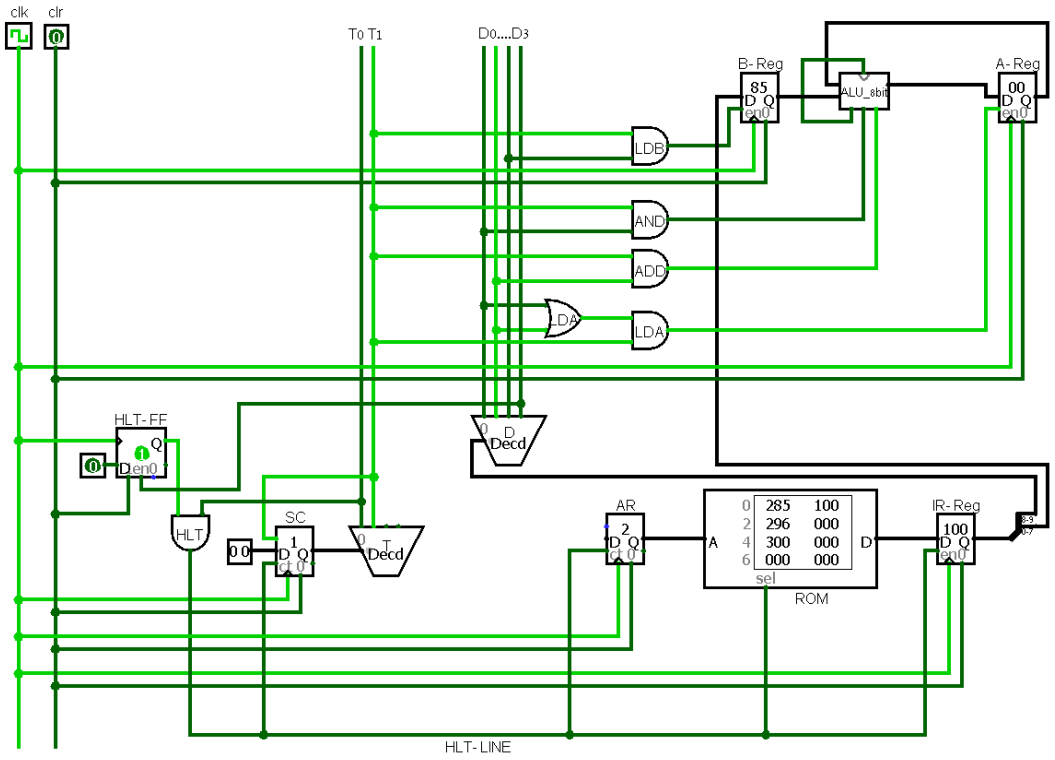
- **Instruction 1: ldb 85h**
  - Fetch (t0): SC & AR increment. Instruction loads into IR and opcode is decoded.



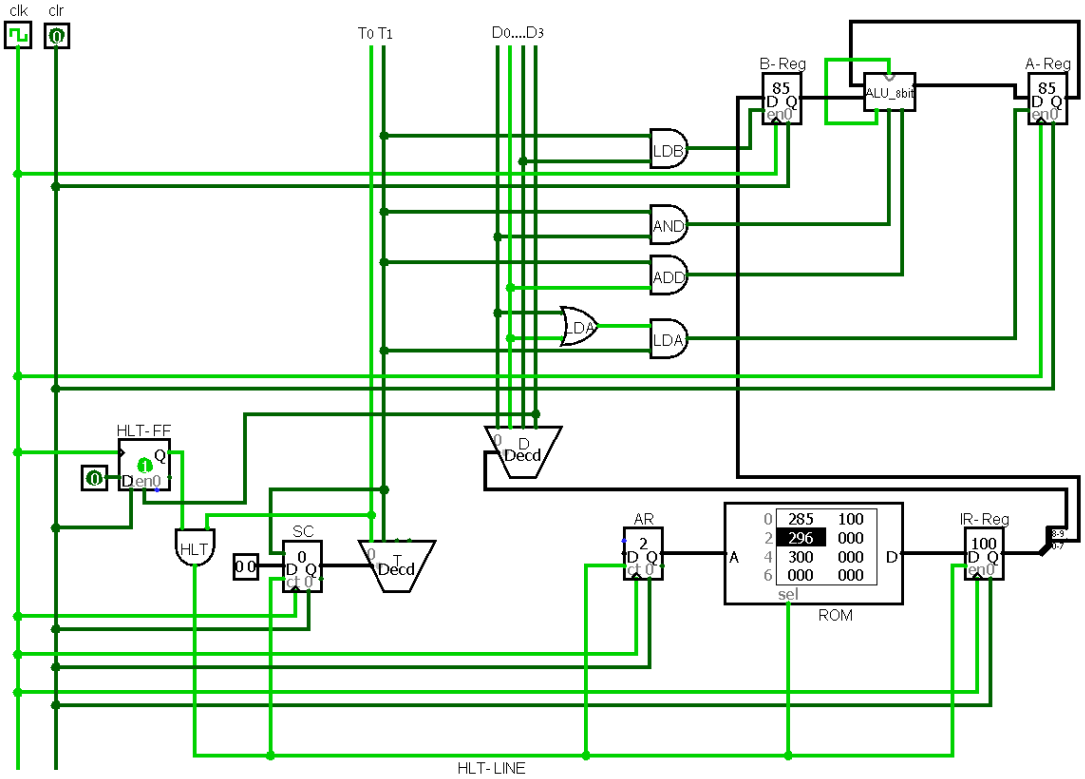
  - Execute (t1): SC resets to 0. The value 85h is loaded into Register B.



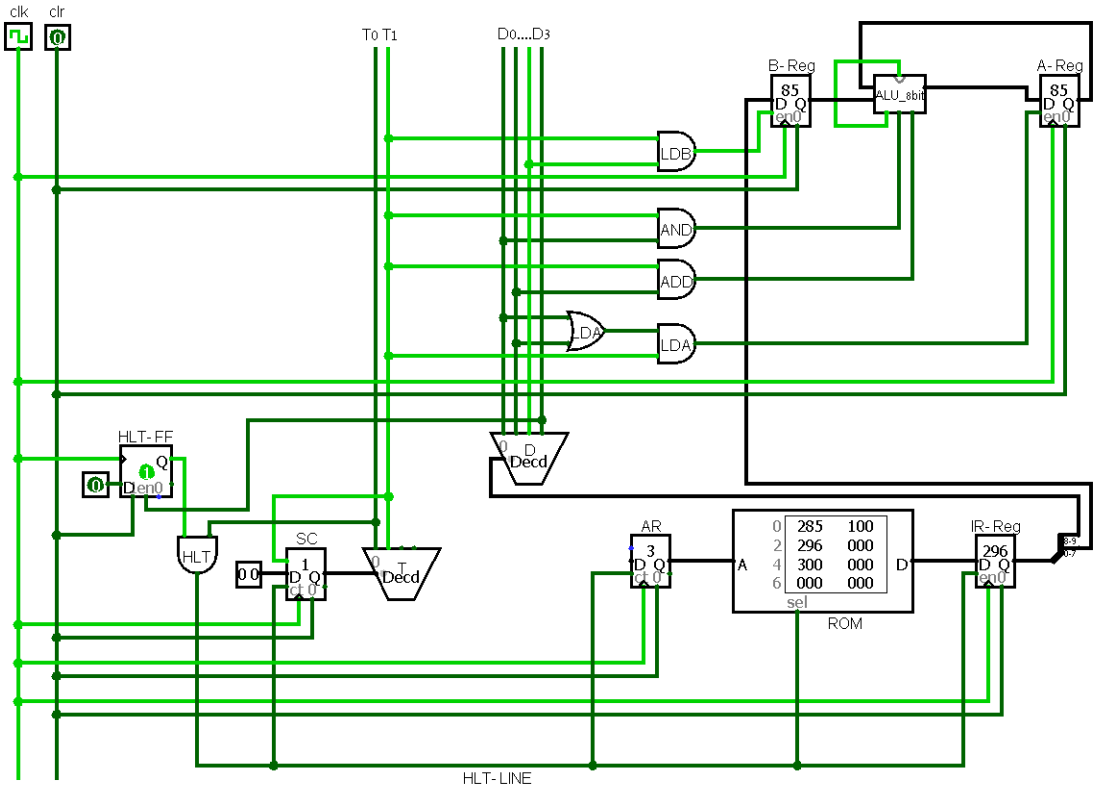
- **Instruction 2: add**
  - Fetch (t0): Standard fetch operation.



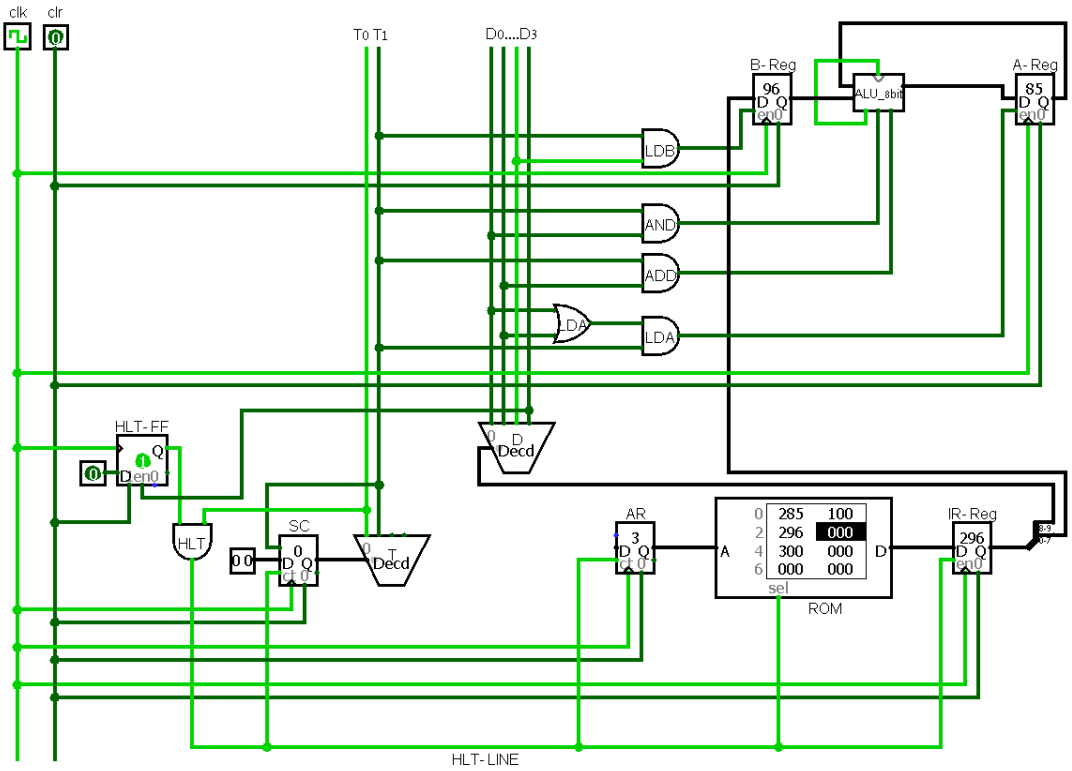
  - Execute (t1): SC resets to 0. ALU adds A and B, storing the result in Register A.



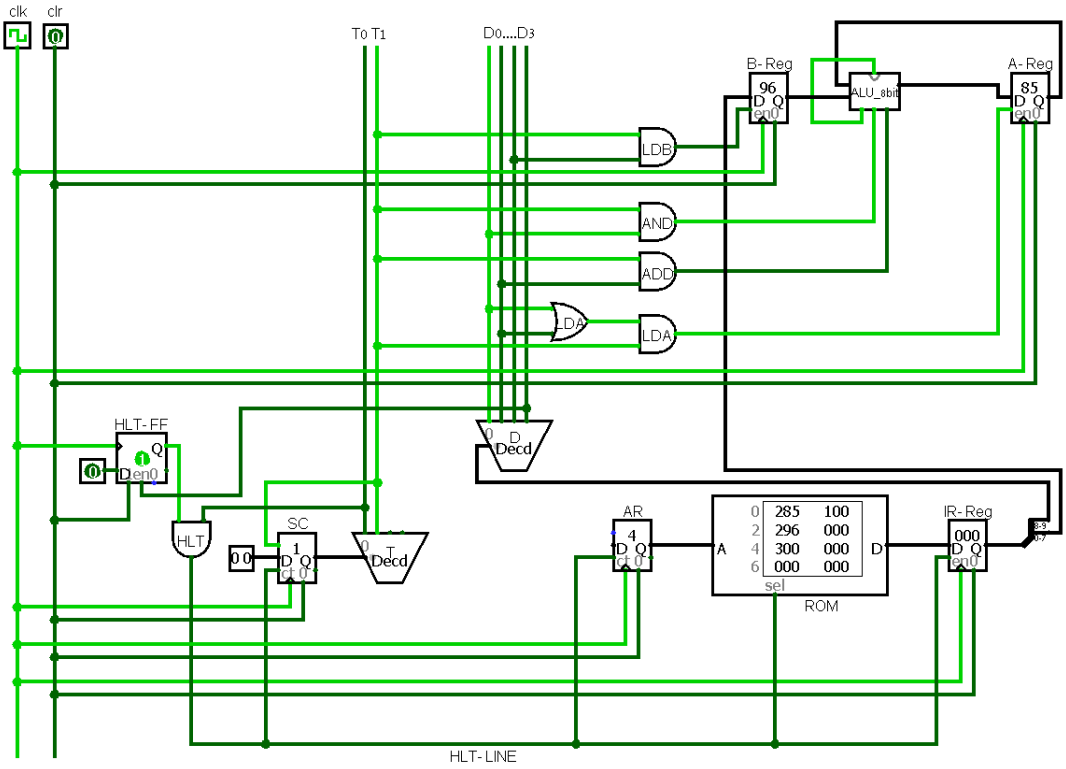
- **Instruction 3: ldb 96h**
  - Fetch (t0): Standard fetch operation.



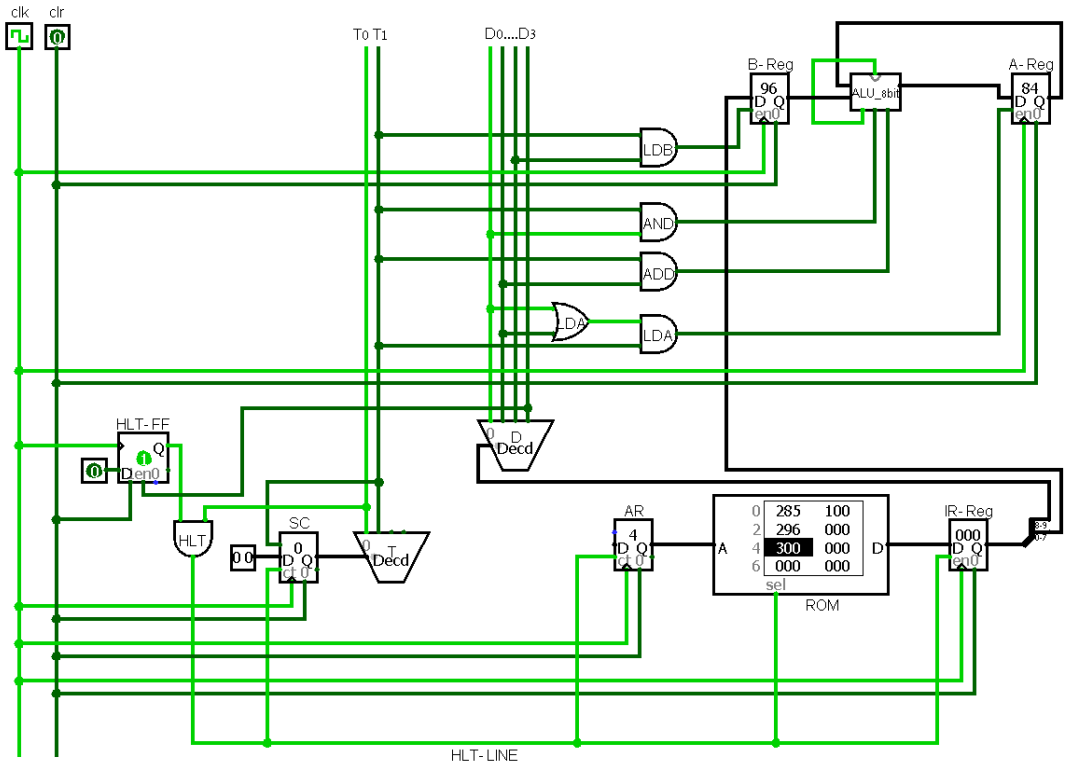
  - Execute (t1): SC resets to 0. The value 96h is loaded into Register B.



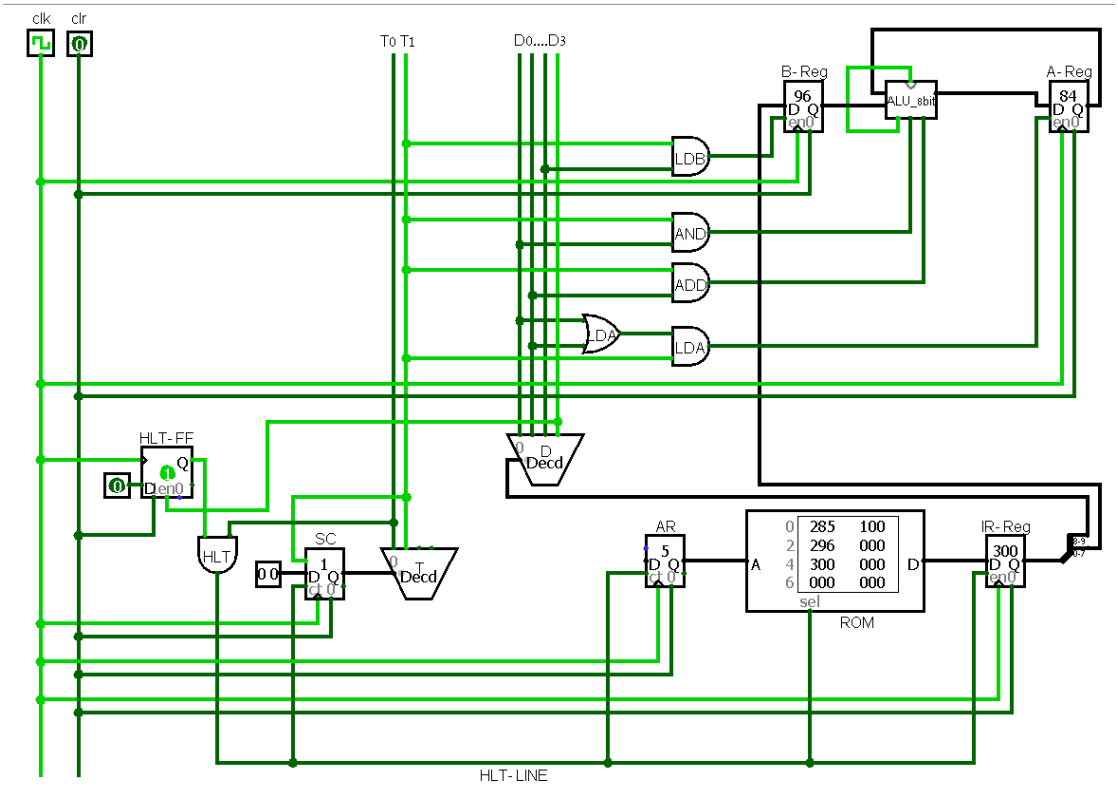
- **Instruction 4: and**
  - Fetch (t0): Standard fetch operation.



  - Execute (t1): SC resets to 0. ALU performs AND on A and B, storing the result in Register A.



- **Instruction 5: hlt**
  - Fetch (t0): Standard fetch operation.



  - Execute (t1): SC resets to 0. HLT Flip-Flop is cleared (0), halting all CPU operations.

